

Network Investigation Report

AgentTesla FTP Data Exfiltration Case

Organization: Globex Manufacturing Ltd.
Analyst Role: SOC Analyst Tier 2
Evidence: 2024-12-04-AgentTesla-variant-using-FTP.pcap
Risk Rating: High

1. Executive Summary

Network traffic analysis confirmed that the internal host 10.12.4.101 was compromised. The host performed DNS lookups for api.ipify.org and ftp.ercolina-usa.com, then authenticated to an external FTP server at 192.254.225.136 using cleartext credentials. The infected host uploaded files containing password data, Chrome cookies, Edge Chromium cookies, and keylogger data using FTP STOR commands. This behavior is consistent with AgentTesla information-stealing malware.

2. Scope and Evidence

Evidence	Description
PCAP file	2024-12-04-AgentTesla-variant-using-FTP.pcap
Primary tools	Wireshark
Primary protocols reviewed	DNS, FTP, FTP-DATA, TCP, HTTPS

3. Infected Host Identification

Item	Finding
Infected host IP	10.12.4.101
FTP server IP	192.254.225.136
DNS server/gateway	10.12.4.1
Evidence	The host 10.12.4.101 communicated with api.ipify.org, resolved ftp.ercolina-usa.com, and uploaded files to 192.254.225.136 over FTP.

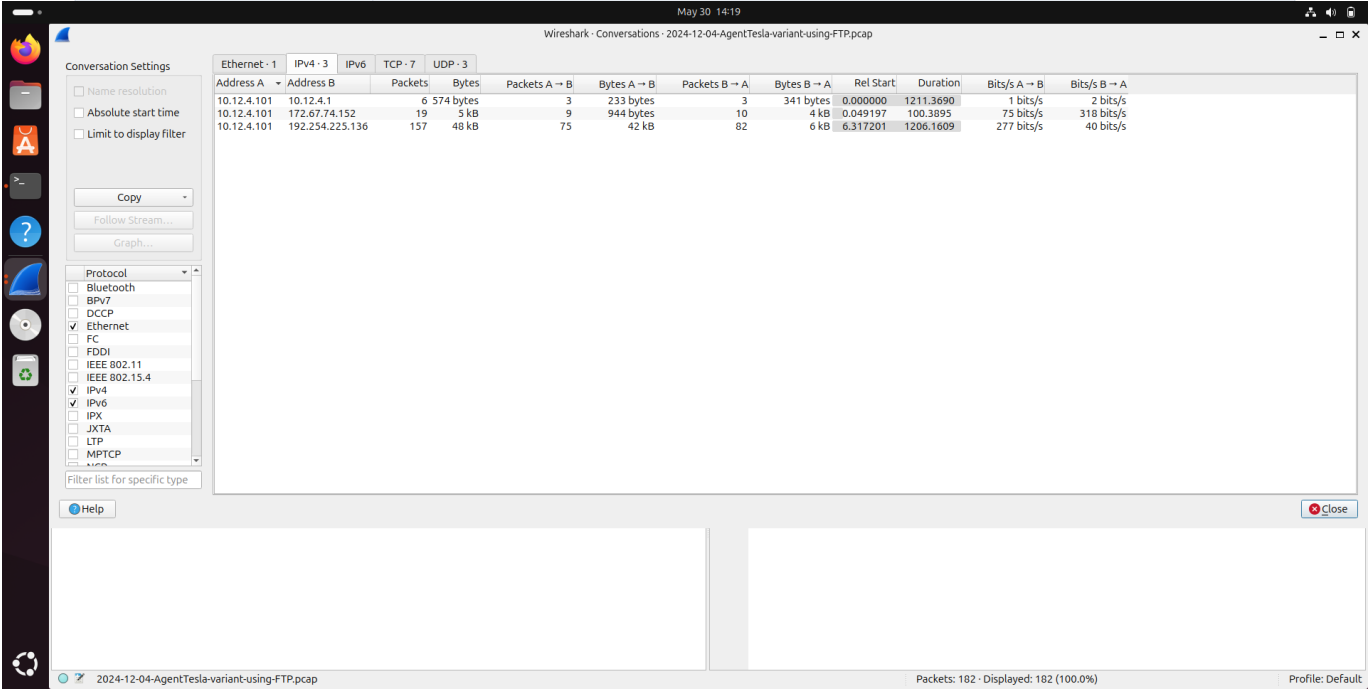


Figure 1: IPv4 conversations showing 10.12.4.101 communicating with external destinations.

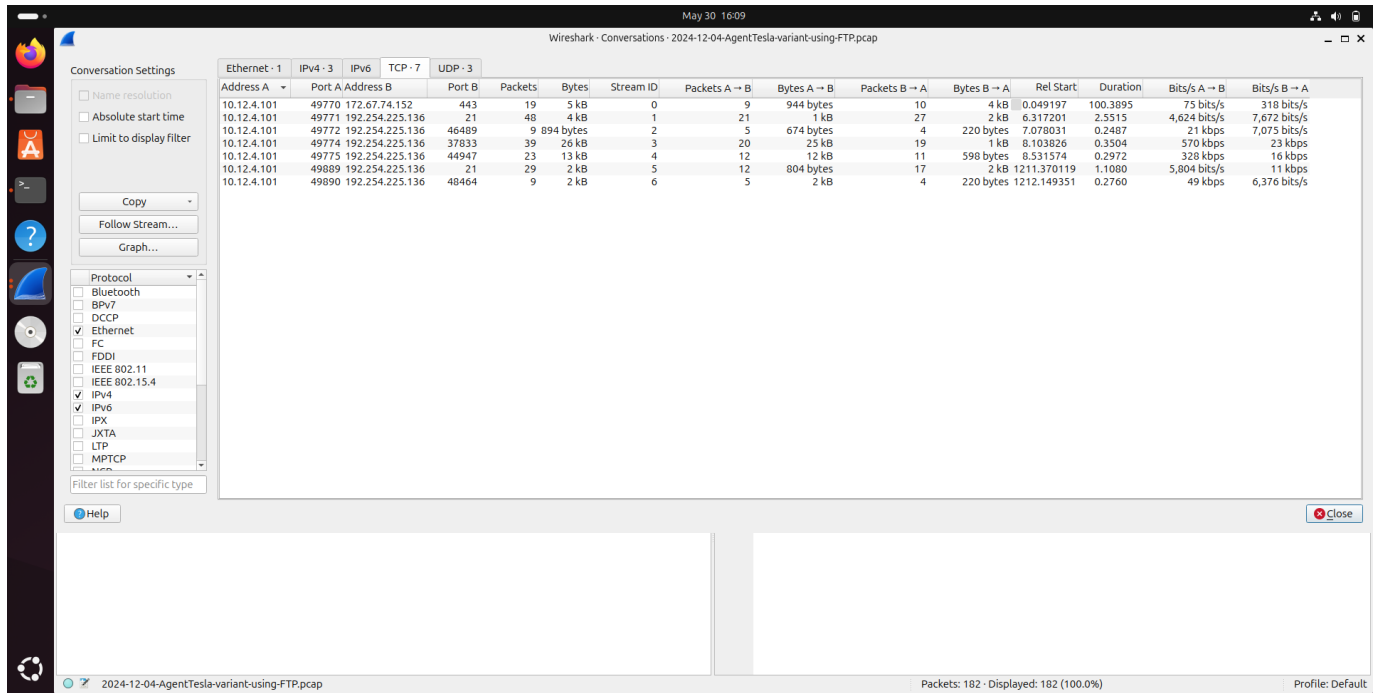


Figure 2: TCP conversations supporting the FTP connection to 192.254.225.136.

4. DNS Analysis

Time	Source	Query	Resolved IP(s)
2024-12-04 21:20:56 UTC	10.12.4.101	api.ipify.org	172.67.74.152, 104.26.12.205, 104.26.13.205
2024-12-04 21:21:02 UTC	10.12.4.101	ftp.ercolina-usa.com	192.254.225.136
2024-12-04 21:41:07 UTC	10.12.4.101	ftp.ercolina-usa.com	192.254.225.136

The api.ipify.org query likely allowed the malware to determine the victim's public IP address. The ftp.ercolina-usa.com query resolved to 192.254.225.136, which was used immediately afterward as the FTP exfiltration destination.

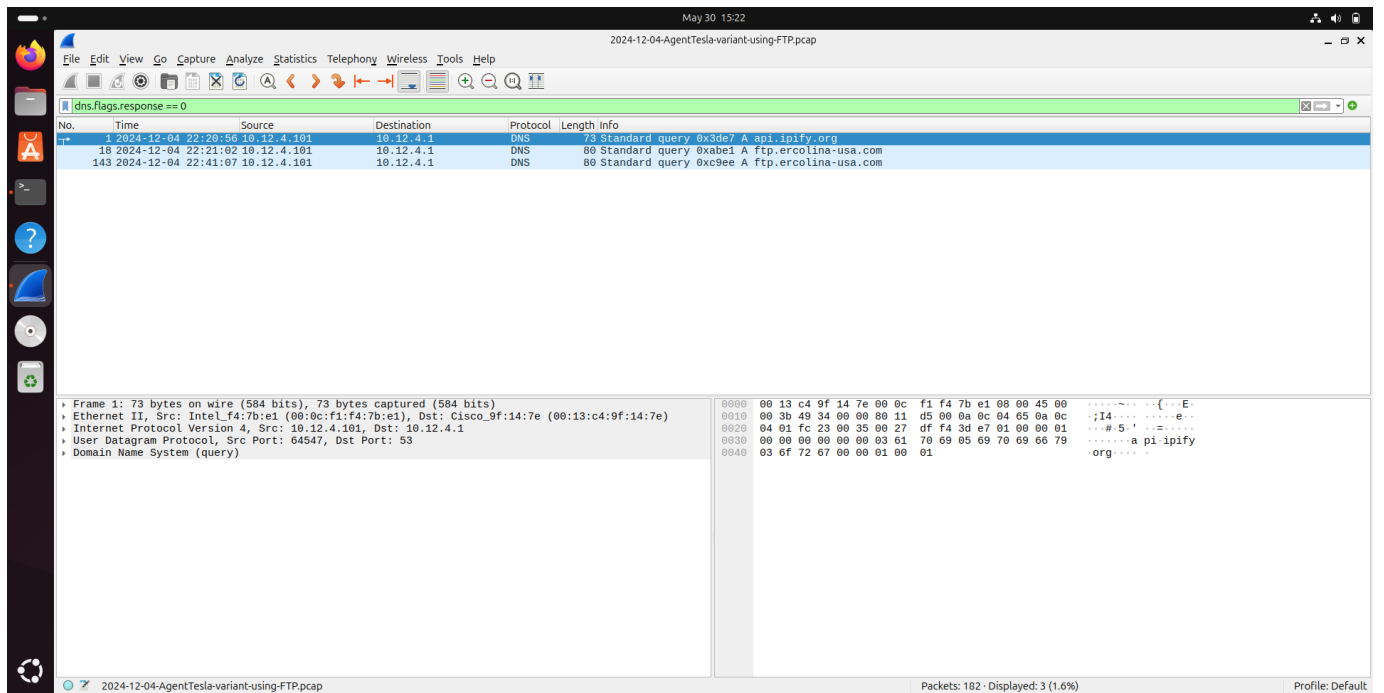
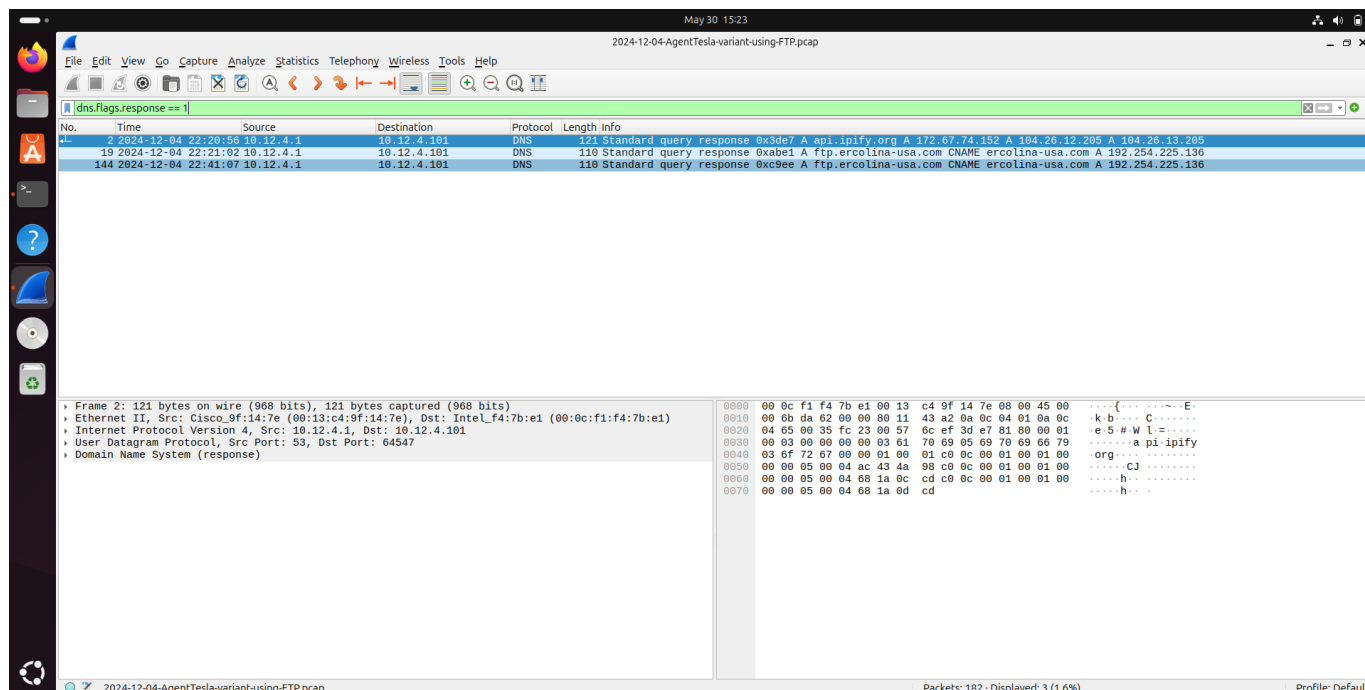


Figure 3: DNS queries from infected host 10.12.4.101.



5. FTP Exfiltration Analysis

Item	Finding
FTP server	192.254.225.136 / ftp.ercolina-usa.com
FTP banner	Pure-FTPD
FTP username	ben@ercolina-usa.com
FTP password	nXe0M~WkW&nJ
Login result	230 OK - login successful
Transfer result	226 - file successfully transferred
Conclusion	FTP was used for data exfiltration. Credentials and files were transmitted without encryption.

Uploaded Files

Time	Uploaded File	Likely Data Type
2024-12-04 21:21:03 UTC	PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html	Passwords / saved credentials
2024-12-04 21:21:04 UTC	CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt	Chrome cookies
2024-12-04 21:21:04 UTC	CO_Edge Chromium_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_04.txt	Edge Chromium cookies
2024-12-04 21:41:08 UTC	KL_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_41_04.html	Keylogger logs

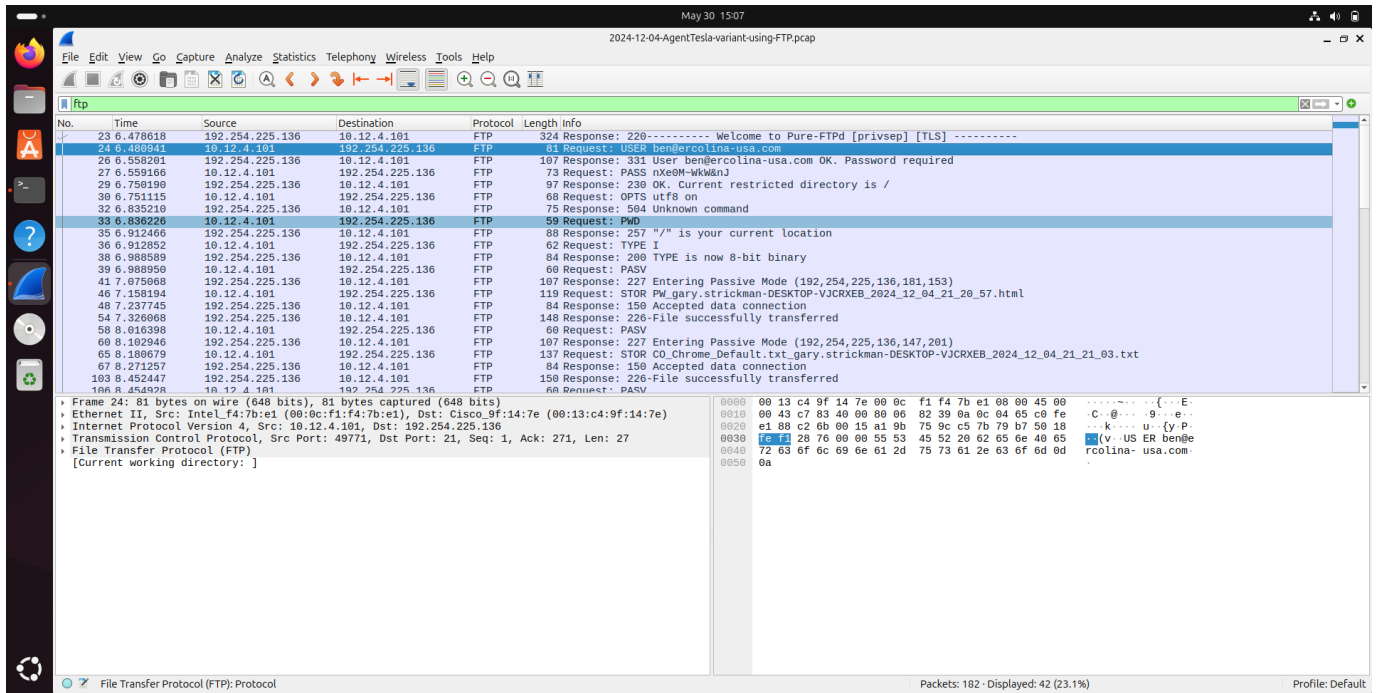


Figure 5: FTP overview showing USER/PASS/STOR commands and server responses.

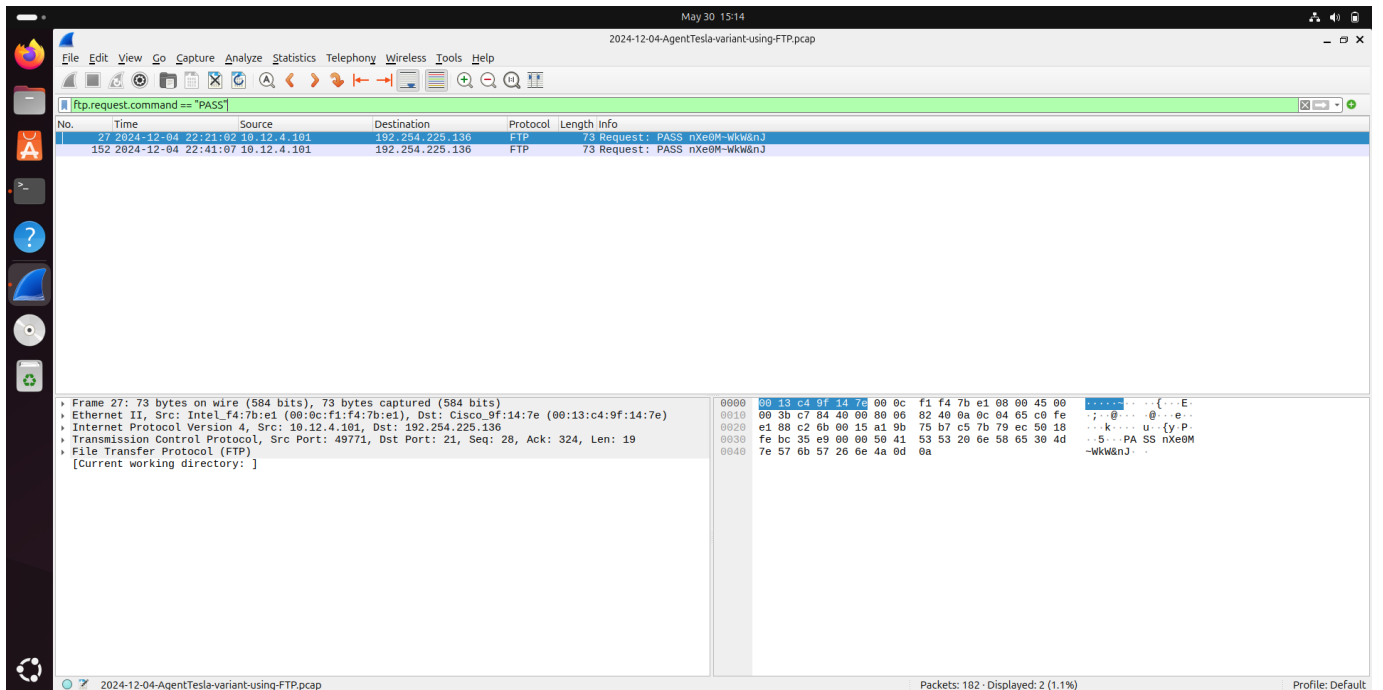


Figure 6: FTP PASS command showing cleartext password transmission.

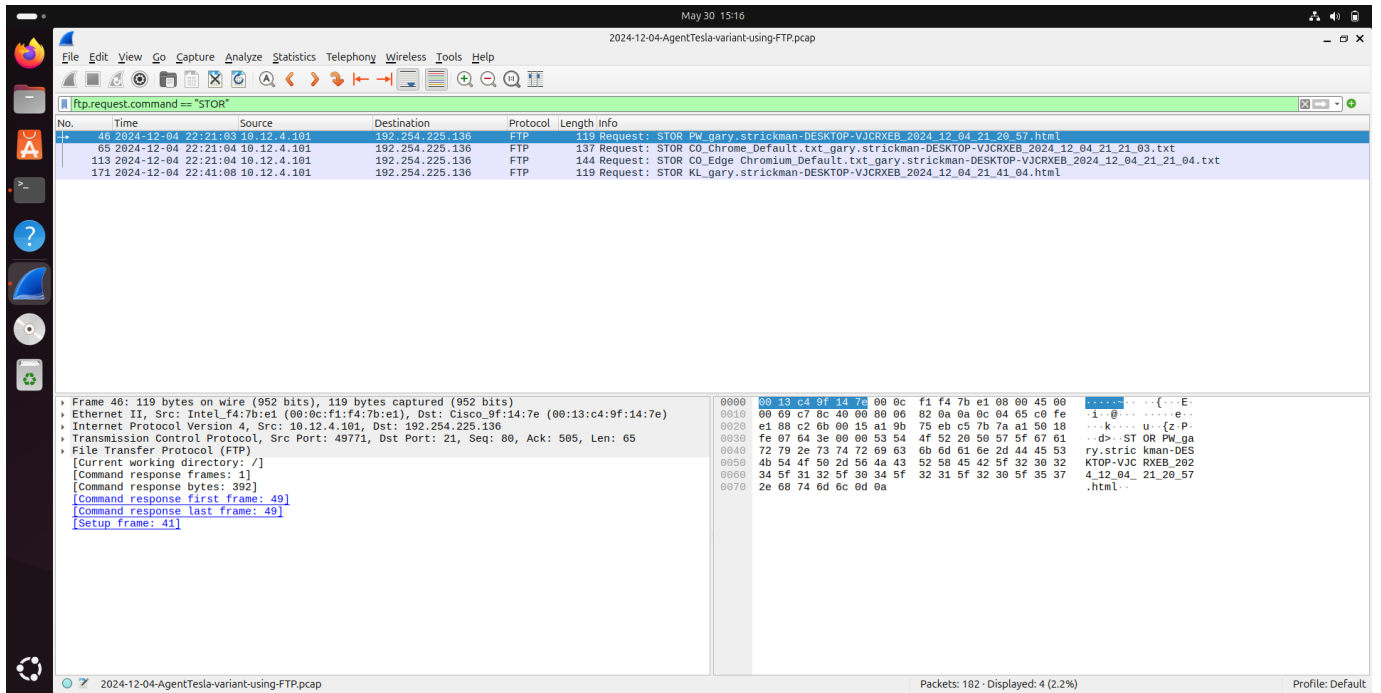


Figure 7: FTP STOR commands showing uploaded files.

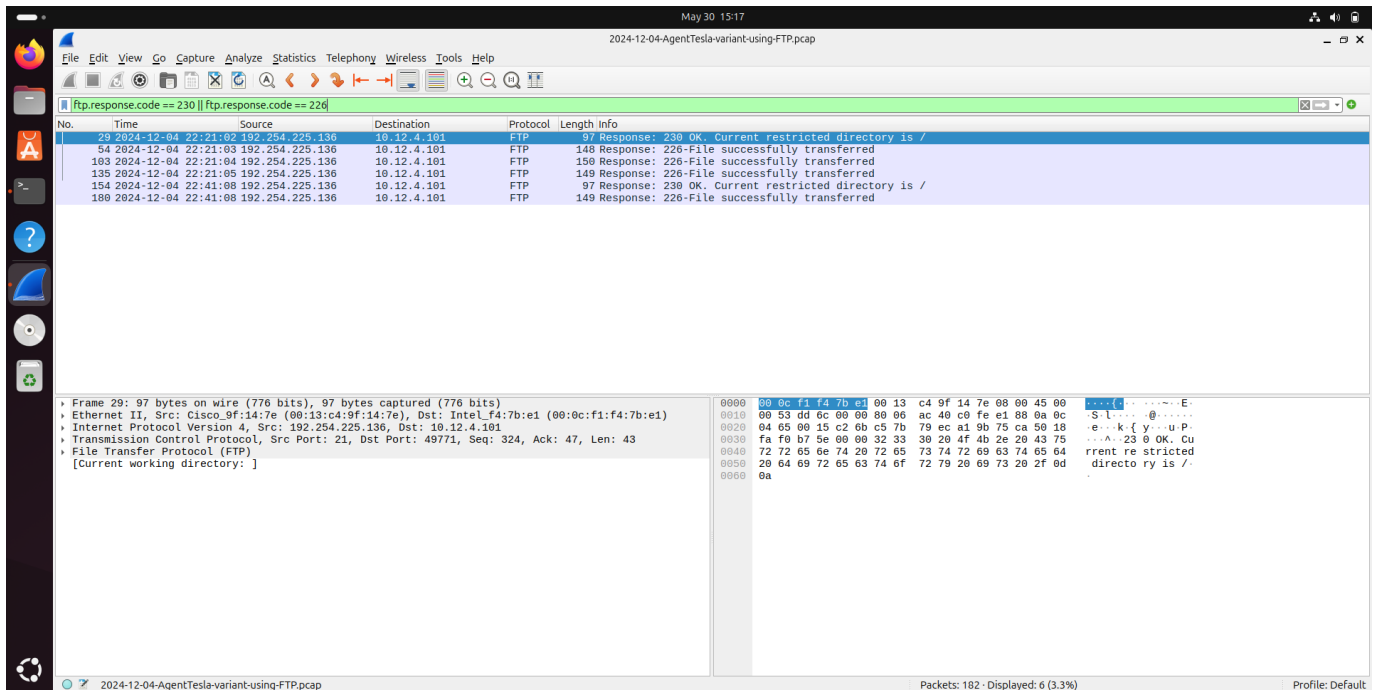


Figure 8: FTP 230/226 response codes confirming login and successful file transfers.

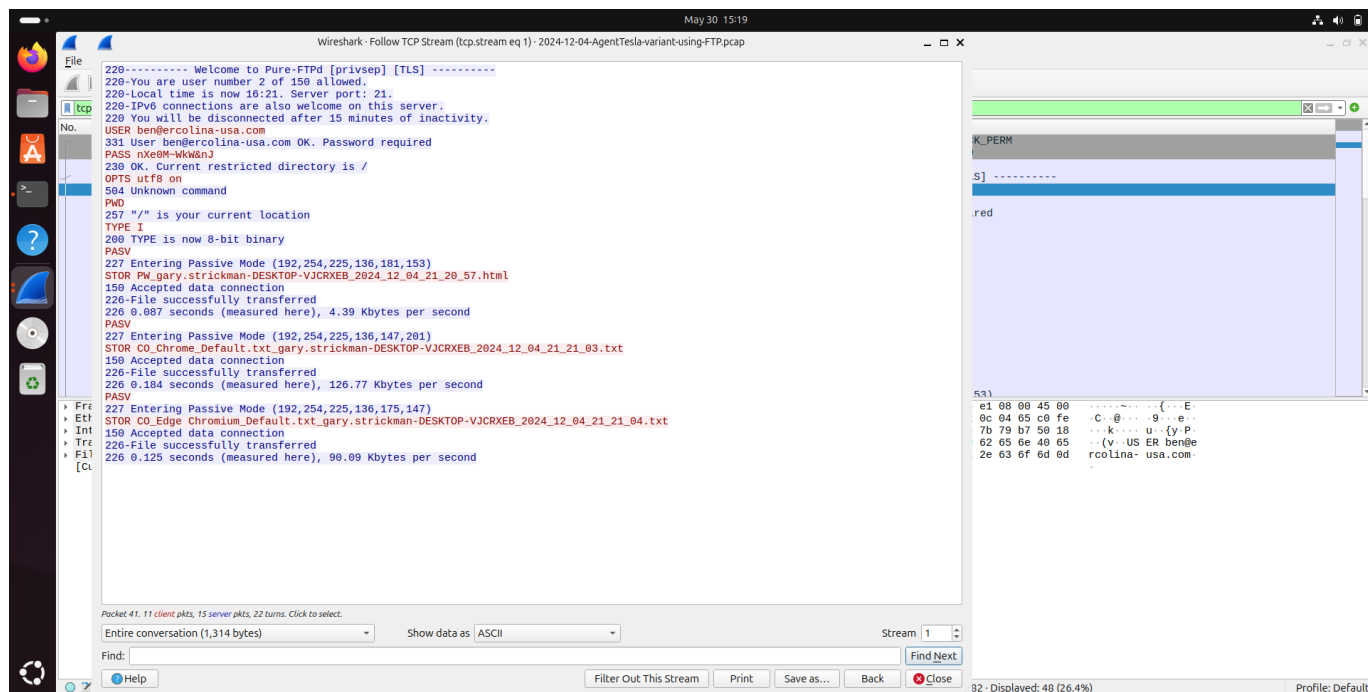


Figure 9: Follow TCP Stream showing FTP login and file uploads.

6. Attack Timeline

Time (UTC)	Event	Evidence
2024-12-04 12:51:17	Phishing email sent	Email Date header: 4 Dec 2024 04:51:17 -0800
2024-12-04 21:20:56	Infected host queried api.ipify.org	DNS query from 10.12.4.101
2024-12-04 21:21:02	Infected host queried ftp.ercolina-usa.com	DNS query from 10.12.4.101
2024-12-04 21:21:02	FTP domain resolved	ftp.ercolina-usa.com resolved to 192.254.225.136
2024-12-04 21:21:02	FTP login initiated	USER ben@ercolina-usa.com
2024-12-04 21:21:02	FTP password sent	PASS nXe0M~WkW&nJ
2024-12-04 21:21:02	FTP login successful	Server response 230 OK
2024-12-04 21:21:03	Password data uploaded	STOR PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html
2024-12-04 21:21:04	Chrome cookies uploaded	STOR CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt
2024-12-04 21:21:04	Edge Chromium cookies uploaded	STOR CO_Edge_Chromium_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_04.txt
2024-12-04 21:41:08	Keylogger data uploaded	STOR KL_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_41_04.html

7. Network Indicators of Compromise

IOC Type	Value	Description
Infected Host IP	10.12.4.101	Internal compromised workstation
DNS Server / Gateway	10.12.4.1	Internal DNS/gateway observed in PCAP
FTP Server IP	192.254.225.136	External FTP exfiltration server
FTP Domain	ftp.ercolina-usa.com	FTP domain resolved before exfiltration
Domain	api.ipify.org	Used to identify public IP address
api.ipify.org IP	172.67.74.152	Resolved IP in DNS response
api.ipify.org IP	104.26.12.205	Resolved IP in DNS response
api.ipify.org IP	104.26.13.205	Resolved IP in DNS response
FTP Username	ben@ercolina-usa.com	Cleartext FTP username
FTP Password	nXe0M~WkW&nJ	Cleartext FTP password

IOC Type	Value	Description
Uploaded File	PW_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_20_57.html	Password data upload
Uploaded File	CO_Chrome_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_03.txt	Chrome cookie data upload
Uploaded File	CO_Edge Chromium_Default.txt_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_21_04.txt	Edge Chromium cookie data upload
Uploaded File	KL_gary.strickman-DESKTOP-VJCRXEB_2024_12_04_21_41_04.html	Keylogger data upload

8. Protocol Overview

The Protocol Hierarchy screenshot provides a high-level view of the protocols observed in the PCAP and supports the presence of DNS and FTP traffic during the investigation.

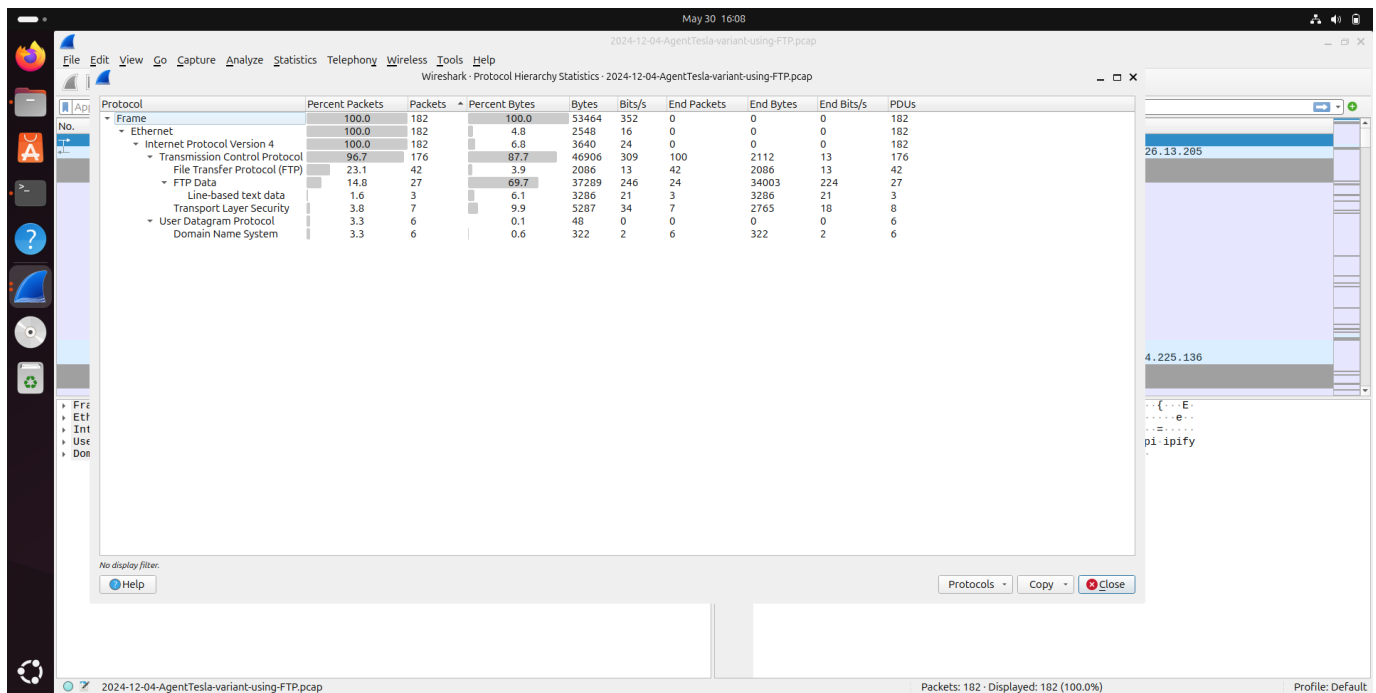


Figure 10: Protocol hierarchy from Wireshark.

9. Conclusion and Recommendations

The PCAP confirms compromise and data exfiltration. The infected host 10.12.4.101 authenticated to the FTP server 192.254.225.136 with cleartext credentials and uploaded multiple files containing passwords, cookies and keylogger data. The incident is assessed as High risk. Recommended actions include isolating the host, resetting affected credentials, blocking the FTP server IP/domain, restricting outbound FTP, hunting for the same IOCs across the environment, and strengthening email attachment filtering and user awareness.